

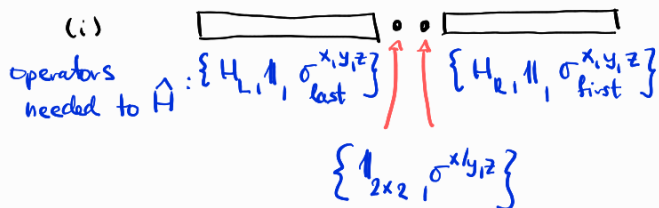
Class 7

Formulation of DMRG with Matrix Product Operators

Example: XXZ-model

$$H = \sum_i \left(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \cdot \sigma_i^z \sigma_{i+1}^z \right)$$

DMRG-step:



(ii) • formation of

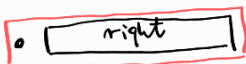


$$H_L^{(l+1)} = H_L^{(l)} \otimes 1_{2 \times 2} + \sum \sigma_{last}^{x(l)} \otimes \sigma^x + \sum \sigma_{last}^{y(l)} \otimes \sigma^y + \sum \Delta \sigma_{last}^{z(l)} \otimes \sigma^z$$

$$1^{(l+1)} = 1^{(l)} \otimes 1_{2 \times 2}$$

$$\sigma_{last}^{x,y,z(l+1)} = 1^{(l)} \otimes \sigma^{x,y,z}$$

• formation of



$$H_R^{(r+1)} = 1_{2 \times 2} \otimes H_R^{(r)} + \sum \sigma^x \otimes \sigma_{first}^{x(r)} + \sum \sigma^y \otimes \sigma_{first}^{y(r)} + \sum \Delta \sigma^z \otimes \sigma_{first}^{z(r)}$$

$$1^{(r+1)} = 1_{2 \times 2} \otimes 1^{(r)}$$

$$\sigma_{first}^{x,y,z(r+1)} = \sigma^{x,y,z} \otimes 1^{(r)}$$

(iii) diagonalization of $\hat{H}$

$$\hat{H} = H_L^{(l+1)} \otimes 1^{(r+1)} + 1^{(l+1)} \otimes H_R^{(r+1)} + \sum \sigma_{last}^{x(l+1)} \otimes \sigma_{first}^{x(r+1)} + \sum \sigma_{last}^{y(l+1)} \otimes \sigma_{first}^{y(r+1)} + \sum \Delta \sigma_{last}^{z(l+1)} \otimes \sigma_{first}^{z(r+1)}$$

(iv)

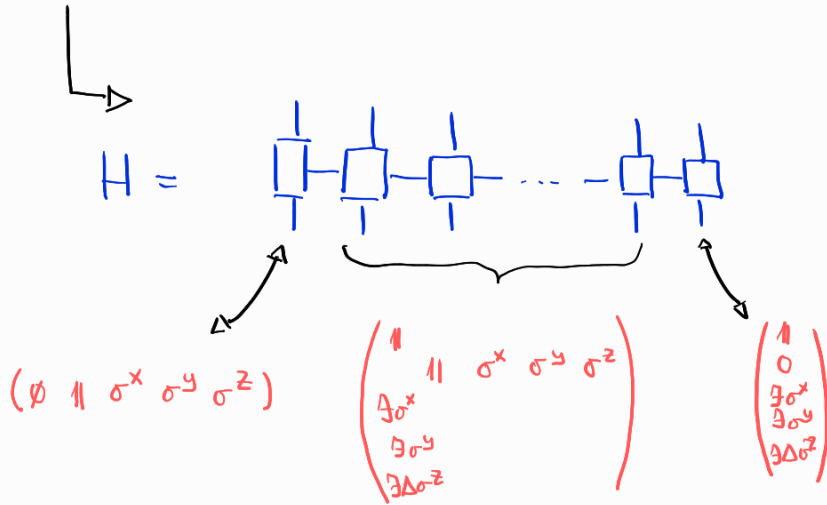
$$|\psi\rangle \rightarrow \sum_{\alpha} \lambda_{\alpha} |\alpha\rangle_{l+1} |\alpha\rangle_{r+1} \rightarrow \text{truncation} \quad \left(\begin{matrix} U_{i\alpha} \\ V_{j\beta} \end{matrix} \right)$$

↳ renormalization of $(l+1)$ ops with U and $(r+1)$ ops with V

Structure of the left update rules:

$$\begin{pmatrix} 1 & & & & \\ & 1 & & & \\ \partial\sigma^x & & \sigma^x & \sigma^y & \sigma^z \\ \partial\sigma^y & & & & \\ \partial\Delta\sigma^z & & & & \end{pmatrix}$$

$$(H_L, 1, \sigma_{\text{last}}^x, \sigma_{\text{last}}^y, \sigma_{\text{last}}^z)_{l+1} = (H_L, 1, \sigma_{\text{last}}^x, \sigma_{\text{last}}^y, \sigma_{\text{last}}^z)_l \otimes$$



Construction based on the right updates

$$\begin{pmatrix} 1 \\ H_R \\ \sigma_{\text{first}}^x \\ \sigma_{\text{first}}^y \\ \sigma_{\text{first}}^z \end{pmatrix}_{n+1} = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ \sigma^x & & \partial\sigma^x & \partial\sigma^y & \partial\Delta\sigma^z \\ \sigma^y & & & & \\ \sigma^z & & & & \end{pmatrix} \otimes \begin{pmatrix} 1 \\ H_R \\ \sigma_{\text{first}}^x \\ \sigma_{\text{first}}^y \\ \sigma_{\text{first}}^z \end{pmatrix}_n$$

almost the same



origin of the difference

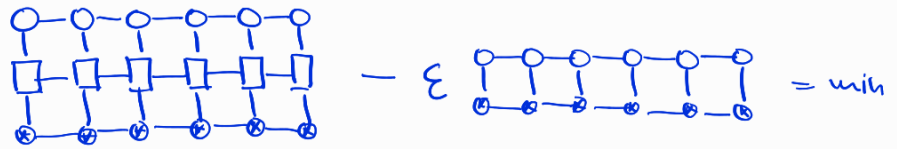
$$H = H_L \otimes 1 + 1 \otimes H_R + \sigma_{\text{last}}^x \otimes (\partial\sigma_{\text{first}}^x) + \sigma_{\text{last}}^y \otimes (\partial\sigma_{\text{first}}^y) + \sigma_{\text{last}}^z \otimes (\partial\Delta\sigma_{\text{first}}^z)$$

$$\text{or } H = H_L \otimes 1 + 1 \otimes H_R + (\partial\sigma_{\text{last}}^x) \otimes \sigma_{\text{first}}^x + (\partial\sigma_{\text{last}}^y) \otimes \sigma_{\text{first}}^y + (\partial\Delta\sigma_{\text{last}}^z) \otimes \sigma_{\text{first}}^z$$

DMRG with MPO

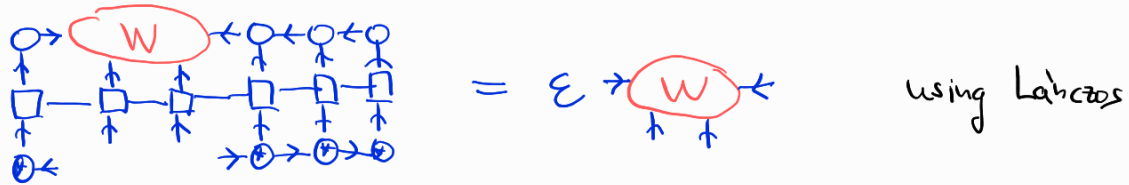
$$\langle \psi | H | \psi \rangle = \min$$

$$\langle \psi | \psi \rangle = 1$$

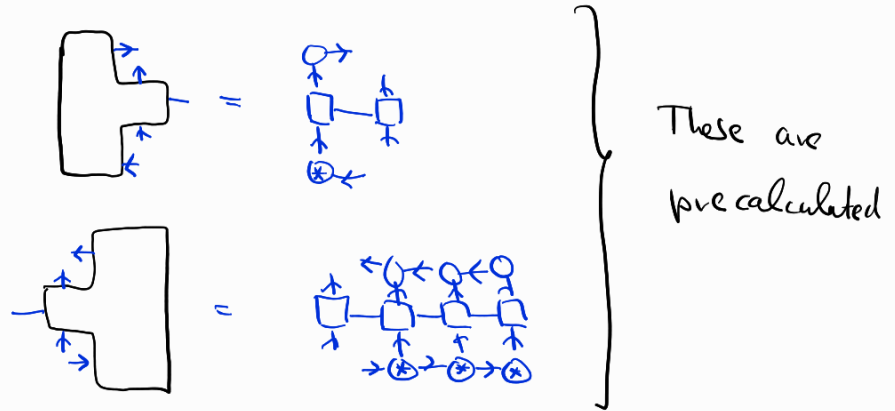


i) optimize two neighboring matrices

- use left canonical tensors in the left and right canonical in the right!

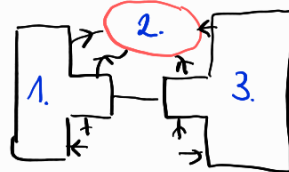


Remark : $(l+1) - (r+1)$ solver



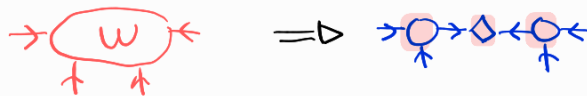
Cost of $H|\psi\rangle$

$$2 \cdot M_{\text{MPO}} (M.d)^3$$



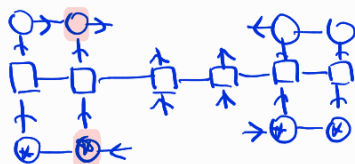
contraction order : $(1 \times 2) \times 3$

ii) Schmidt decomp



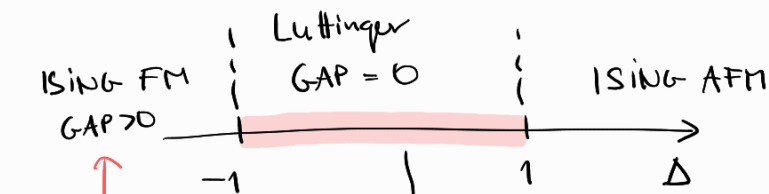
iii) 1 step right or left and GoTo i)

(sweeping)



in infinite chain case, we keep both U & V and plug in 2 new sites

Numerical DEMO: XXZ-model



↑↑↑↑↑
exact ground state

$$\xi = \infty$$

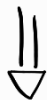


$$S_{\text{segment}}^l \sim \log(l)$$

Cable-knot - Gandy

$$S_{\text{left half}}^L \sim \log \frac{L}{2}$$

required M increases
polynomially



$$S^l \sim \log l \ll \sim l$$

(Volume law)

DMRG is still much better than direct ED!

$$\text{GAP} > 0$$



$$\xi < \infty \quad \text{correlation length}$$



Area law entanglement

$$S_{\text{left}}^{L \rightarrow \infty} = \text{finite}$$



finite M is enough
in the TDL



That's why DMRG
performs excellently
for gapped local
1D chains

Fermion models

Hubbard chain

$$H = -t \sum_{i,\sigma} (c_{i\sigma}^\dagger c_{i+1\sigma} + c_{i+1\sigma}^\dagger c_{i\sigma}) + u \sum_i n_{i\uparrow} n_{i\downarrow}$$

1-site ops:

$$C_{\uparrow}^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$u_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

$$C_{\downarrow}^{\dagger} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$

$$h_{\downarrow} = \begin{pmatrix} 0 & & \\ & 1 & \\ & & 0 & \\ & & & 1 \end{pmatrix}$$

$$\Phi = (-1)^{u_{\uparrow} + u_{\downarrow}} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}$$

Jordan-Wigner

$$C_{i\sigma}^\dagger = \phi \otimes \phi \otimes \dots \otimes C_\sigma^\dagger \otimes 1 \otimes \dots \otimes 1$$

$$n_{i\sigma} = 1 \otimes 1 \otimes \dots \otimes n_{\sigma} \otimes 1 \otimes \dots \otimes 1$$

Hopping term

$$C_{i\sigma}^+ C_{i+1\sigma} = 1 \otimes 1 \otimes \dots \otimes C_\sigma^\dagger \otimes C_\sigma \otimes 1 \otimes \dots \otimes 1$$

$$C_{i+1\sigma}^\dagger C_{i\sigma} = \mathbb{1} \otimes \mathbb{1} \otimes \dots \otimes \Phi C_\sigma \otimes C_\sigma^\dagger \otimes \mathbb{1} \otimes \dots$$

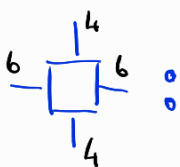
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$$-C_{i\sigma} C_{j+i\sigma}^\dagger = - \cancel{1} \otimes \cancel{1} \sim \otimes C_\sigma \otimes C_\sigma^\dagger \otimes \cancel{1} \dots$$

Left ops to keep

$$(H_L, \mathbb{1}, C_{\text{last} \uparrow}^+, C_{\text{last} \uparrow}, C_{\text{last} \downarrow}^+, C_{\text{last} \downarrow})$$

MPO-matrix:



11	11	$c_{\uparrow}^+$	$c_{\uparrow}$	$c_{\downarrow}^+$	$c_{\downarrow}$
$u_{\uparrow}u_{\downarrow}$					
$-tc_{\uparrow}$					
$tc_{\uparrow}^+$					
$-tc_{\downarrow}$					
$tc_{\downarrow}^+$					

note: sparsity!

Long-range models

spinless fermions with long-range hopping

$$H = \sum_{ij} T_{ij} c_i^\dagger c_j$$

left-right cut

$$H = \overbrace{\left(\sum_{i,j \in \text{left}} T_{ij} c_i^\dagger c_j \right)}^{H_L} \otimes \mathbb{1} + \mathbb{1} \otimes \overbrace{\left(\sum_{i,j \in \text{right}} T_{ij} c_i^\dagger c_j \right)}^{H_R} +$$

$$+ \sum_{i \in \text{left}} \sum_{j \in \text{right}} T_{ij} c_i^\dagger \phi \otimes c_j +$$

$$+ \sum_{i \in \text{right}} \sum_{j \in \text{left}} -T_{ij} c_j \phi \otimes c_i^\dagger$$

Pre-summed operators:

if left > right

$$[Tc^\dagger]_j^{(L)} = \sum_{i \in \text{left}} T_{ij} c_i^\dagger \phi \quad j \in \text{right}$$

$$[Tc]_j^{(L)} = \sum_{i \in \text{left}} -T_{ji} c_i \phi$$

$$H = H_L \otimes \mathbb{1} + \mathbb{1} \otimes H_R + \sum_{j \in \text{right}} [Tc^\dagger]_j^{(L)} \otimes c_j + \sum_{j \in \text{right}} [Tc]_j^{(L)} \otimes c_j^\dagger$$

$$\hookrightarrow \text{MPO bond dimension} : \lesssim 2 + \frac{N}{2}$$

2-body model

$$H = \sum_{ij} T_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$$

Pre-summed ops:

$$[Vc^\dagger c^\dagger]$$

$$[Vc^\dagger c^\dagger + Tc^\dagger]$$

$$[Vc^\dagger c]$$

$$[Vc^\dagger c + Tc]$$

$$[Vcc]$$

MPO bond dimension

$$\sim N^2$$